

Frequently Listed, Rarely Examined: A Screened Bibliometric Map of Warming and Cooling Food Claims in Japan

Running title: What predicts research attention to warm/cool food claims in Japan

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Abstract

Background: Japanese *onkatsu* (“warming care”) rests on a widely shared belief that particular foods warm or cool the body. Prior reviews find the scientific support heterogeneous rather than absent, but none has asked, food by food, whether the most widely circulated claims are the ones examined.

Methods: Cross-sectional bibliometric study. Axis A counts, per food, how many of 15 Japanese lay-facing sources (frozen in advance; nine corporate/association members form the primary frame) assign it a warming or cooling direction. Axis B counts PubMed hits for each food co-occurring with 22 thermal-effect terms, matched to the outcome classes the screening protocol accepts, then screens every hit, since the raw count measures keyword co-occurrence, not the construct. Paired large language models of adjacent capability tiers from one vendor independently labelled all 12,437 records, with the author adjudicating (Cohen’s $\kappa = 0.806$).

Results: Screening cut 98% of the raw signal: 11,715 hits across 146 foods became 283 on-construct studies, and 97 foods (66.4%) had none. In the planned primary model, adjusted for total PubMed volume (L1), volume predicted study presence (odds ratio [OR] 2.106 per unit of $\log(L1 + 1)$, 95% confidence interval [CI] 1.583–2.803, $p < 0.001$); coverage did not (OR 1.038, 95% CI 0.854–1.263, $p = 0.705$), though it compounds to 0.28–6.47-fold over one-to-nine sources. Unadjusted, coverage was associated with study presence (OR 1.272, 95% CI 1.075–1.504, $p = 0.005$). Chicken, warming in six sources, returned 702 raw hits and five studies, two of which examined people eating chicken meat.

Conclusions: General literature volume was associated with whether a warming/cooling claim had been directly examined, whereas no association with sampled lay-source coverage was detected once volume was held fixed; the wide interval does not exclude a substantial coverage effect. For two-thirds of these foods, the search retrieved no such study.

Keywords: attention gap; bibliometrics; folk nutrition; warming and cooling foods; thermoregulation; abstract screening; Japan

Introduction

The idea that individual foods “warm” or “cool” the body is a durable feature of everyday health culture in Japan, where it underpins the popular practice of *onkatsu* (温活, “warming care”) and lay reasoning about *hiesho* (冷え症, habitual cold sensitivity). Corporate wellness media, food manufacturers, and individual practitioners routinely publish lists that sort ginger, carrot, and burdock as “warming” and cucumber, tomato, and coffee as “cooling,” and these attributions circulate widely enough to shape ordinary purchasing and seasonal eating advice. That the belief is culturally established, rather than niche, is visible both in the volume of lay-facing sources that maintain such lists and in the fact that Japanese epidemiology has taken the classification seriously: Nagata et al. (2017), analyzing the Takayama cohort ($n = 28,356$), applied four independent warm/cool food-classification lists to real dietary data. The premise of the belief is therefore neither obscure nor recent.

Crucially, the scientific literature on warm/cool food theory is not empty, and this study does not claim that it is. Ormsby (2021), in a scoping review of the nutritional evidence, characterized the support as “heterogeneous and of mixed quality” while identifying partial mechanistic correlates — heating foods associated with higher caloric density, sympathetic activation, and vasodilation, and cooling foods with higher water and fiber content and anti-inflammatory processes. Namiranian et al. (2021), a companion review in the same volume, surveyed the physiological basis of hot/cold theory across Persian medicine, traditional Chinese medicine, and Ayurveda. Zhou & Xu (2021) integrated candidate molecular mechanisms (for example, differential nuclear factor kappa B [NF- κ B] and mitogen-activated protein kinase [MAPK] signaling) linking the cold/hot nature of foods to biological effects — while explicitly noting that research applying these mechanisms to foods is scarce relative to research on Chinese medicines. Any framing of the field as an “evidential desert” is thus untenable; the appropriate description is fragmentary, unintegrated, and unevenly attended.

At the level of individual foods, the available evidence is not only fragmentary but sometimes contradictory or opposite to belief. Ginger, the archetypal “warming” food, showed a thermic-effect difference at $p = 0.049$ in one small randomized crossover pilot — though total resting energy expenditure and the thermic-effect area under the curve were both null in that same trial (Mansour et al. 2012, $n = 10$ overweight men) — while a larger, double-blind crossover trial in a different population reported no increase at all (Fagundes et al. 2021, $n = 20$ normal-weight women). Coffee, popularly classified as “cooling” (or “yin”) in Japan, is dominated by caffeine, whose best-powered synthesis under thermal stress — a meta-analysis pooling $n = 30$ caffeine effects — found that caffeine significantly *raises* peak core temperature (Hedges’ $g = 0.44$; Peel et al. 2025). The physiological direction most relevant to

coffee's principal active compound therefore runs opposite to the folk attribution, even after allowing for the distinct measurement context (heat-exposure performance settings rather than everyday thermal sensation).

Methodologically, the practice of mapping a hot–cold belief system bibliometrically has direct precedent. García-Hernández et al. (2023) assembled and classified 101 academic publications spanning roughly a century to characterize Mexico's hot–cold system by research approach, depth, and conceptual domain. Nagata et al. (2017) demonstrated, within Japan, that the four extant warm/cool classification lists disagree with one another when applied to cohort data — an internal inconsistency that is itself a symptom of the field's uneven development. These works establish that belief systems of this kind can be quantified and that the classification schemes underlying them are neither unified nor exhaustively validated.

What remains unquantified is the *asymmetry* between the two facts above: that certain foods are very widely presented as warming or cooling, and that the direct scientific attention paid to those specific claims varies enormously. Ormsby (2021) asked whether mechanistic support exists but did not measure how widely each food is claimed, nor compare research volume across foods; reviews of the adjacent clinical concept of cold hypersensitivity (Jin et al. 2025, 65 studies) consistently take traditional-medicine interventions, not lay food beliefs, as their subject. We therefore ask a descriptive question: across the foods that Japanese lay sources classify as warming or cooling, is the breadth of that lay coverage related to whether the specific thermal claim has been directly studied — and, if there is any apparent relationship, does it survive adjustment for how much the food is studied for any reason at all?

Answering that question turns out to depend on a measurement problem that a keyword count cannot solve, and addressing it is the methodological contribution of this paper. Counting how often a food name co-occurs with thermal vocabulary in PubMed does not count studies of that food's thermal effect in people: the co-occurrence set is dominated by poultry and livestock heat-stress research, and it misses the nutrition trials that report the thermic effect of food. We therefore screened the entire co-occurrence set to the intended construct before analysing it, using two independent language models with author adjudication — a design with recent published precedent in evidence synthesis (Guo et al. 2024; Hilkenmeier et al. 2026).

Methods

Design

This is a descriptive, cross-sectional bibliometric study. It maps two independently constructed axes against each other for a fixed set of foods and reports their association; it does not estimate a causal effect of lay coverage on research or vice versa. Because the study was not registered on any protocol registry, we describe fixed-in-advance choices as *planned* rather than pre-specified.

Axis A: lay-source coverage

Axis A quantifies how many independent Japanese lay-facing sources assign each food a warming or cooling direction, following the bibliographic-coding approach of García-Hernández et al. (2023). We use “coverage” rather than “belief” throughout: the measure counts sources that publish an attribution, and is a proxy for — not a measurement of — how widely the attribution is actually held.

How the frame was assembled. Candidate sources were located by Japanese-language web search in two passes. The first swept general queries for the belief itself — among them 「体を温める食べ物 冷やす食べ物 一覧 漢方」 (“list of foods that warm or cool the body, kampo”), 「温活 食材リスト 温める 冷やす 分類」 (“onkatsu ingredient list, warming/cooling classification”), 「五性 温熱 涼寒 食材 一覧」 (“five-nature ingredient list”), and 「冷え性 食べ物 温める 冷やす 一覧」 (“hiesho food list, warming and cooling”). The second ran domain-restricted queries over the sectors that publish such lists: kampo and pharmaceutical manufacturers, health-food companies, drugstore chains, food and beverage makers, medicinal-food (*yakuzen*) associations, and the web editions of women’s and lifestyle magazines. Two independent existence scans then verified every candidate by fetching the page — publishing about warming care in general was not sufficient; the page had to assign individual foods a direction — and by retrieving its robots.txt. Independence was judged on operator and on content, so aggregator pages republishing another site’s list were dropped. The frame is therefore a purposive sample of what was reachable on the Japanese-language web in August 2026, assembled before any Axis B count existed; it is not a probability sample of Japanese lay media, and it is not weighted by readership.

The sampling frame was frozen in advance: 15 sources verified to be reachable over HTTP, to present a concrete per-food warm/cool list, and to permit fetching per robots.txt. Nine corporate- or association-operated sources form the **primary (Tier 1) frame**; six individual-expert blogs form a **Tier 2 sensitivity frame**, which the coding protocol reserves for testing whether conclusions are stable. Sources sharing a single operator were counted once. Two frame corrections were made during fetching and before any coding of results — one source lacking a concrete per-food list was excluded, and one source’s URL was corrected to its per-food list page — with both decisions driven by page content rather than by any result.

Each source page was fetched once, on 2026-08-03 (one source re-fetched 2026-08-04 at its corrected URL), and stored locally; the per-source access dates are published in `data/sources.csv`. Every food mentioned was then coded for its direction (warming, cooling, or neutral), with the verbatim supporting quotation recorded on each row. Coding rules were documented before coding: sources using traditional five-nature or yin–yang vocabulary were mapped to warming (温性/熱性/陽性), cooling (涼性/寒性/陰性), or neutral (平性); serving temperature (e.g. “cold drinks”) is not a food’s nature and was not coded as one. For each food, Axis A records the number of independent sources assigning a direction (`n_sources`) and the majority direction; a consensus ratio is also computed and published with the data, but no analysis here uses it.

Who coded, and what the agreement statistic can support. All coding was performed by large language models operating under the author’s direction; there were no human raters. Each source was coded twice, independently: the first pass over six Tier-1 sources by Claude Opus 4.8 reading the stored text, the remaining nine sources by separate Claude Sonnet 4.6 agents, and the second pass throughout by further Claude Sonnet 4.6 agents that received only the protocol and the stored source text, with the first coder’s output withheld. Codings were reconciled on the (source, food, condition) key. Cohen’s κ was 1.000 across the two rounds, but on one specific quantity: **the direction assigned to the 481 items both coders extracted**. It is not a reliability estimate for extraction, and extraction is where this task’s disagreement actually lived. The coders returned different numbers of records — in round 2, 236 and 377, agreeing on 214 — so the extraction overlap in that round is a Jaccard index of 0.54, not anything near unity. The 190 coverage differences across the two rounds — five in round 1 and 185 in round 2, a coverage difference being a food one coder recorded and the other did not — were accordingly adjudicated one at a time against the source text. That reading dropped eleven records that carried no nature attribution in the source, added one the first coder had missed, and corrected three misread directions; the remainder were adjudicated and accepted as coded. These are the counts of the reconciliation as executed on 2026-08-04; two food keys were later merged and split (a synonym pair, and a dried-versus-fresh distinction three sources had assigned inconsistently), so re-running the same reconciliation against the frozen dataset returns 218 co-coded items and 177 coverage differences in round 2, with direction agreement unchanged at $\kappa = 1.000$.

That κ pools two differently constituted rounds, and neither supports an independence claim. In round 2 (214 of the 481 co-coded items) both coders were separate instances of the same model, so their agreement measures the stability of one model’s reading. In round 1 (the other 267 items) the two coders were different capability tiers of the same vendor’s line — the same arrangement used for the Axis B screening — whose errors should be expected to correlate. Perfect agreement across that tier boundary is itself evidence of shared behaviour rather than of independent convergence. The κ is therefore an upper-bound-leaning consistency estimate, not comparable to a κ between human raters; we return to this in the Limitations. The reproducibility evidence we rely on instead is deterministic: `verify_claims.py` checks that every retained row occurs verbatim in the stored source text, and every row carries its source and quotation, so the coding can be re-derived from the published frame rather than taken on trust.

Coding yielded 649 rows and 196 distinct foods. Of these, 157 are covered by at least one Tier-1 source; 11 are composite category labels (“nuts,” “spices,” “shellfish,” “leafy greens” and similar) that cannot be turned into a single unambiguous search query and were excluded from Axis B, leaving **146 foods** in the primary analysis. A normative five-nature classification, against which lay attribution could be benchmarked, was planned but is deferred to future work because a citable primary source could not be secured within scope.

Axis B: on-construct research attention

Raw counts. Axis B begins from the National Center for Biotechnology Information (NCBI) PubMed E-utilities (no API key; queries rate-limited under 3 requests/s). All PubMed, CiNii, and OpenAlex queries were executed on 2026-08-06 to 2026-08-07; because bibliometric counts grow over time, this snapshot fixes the reported values. For each food we counted hits at two nested layers: **L1**, the food term alone, as total research volume; and **L2**, the food term co-occurring with any of **22 thermal-effect terms**. The vocabulary is set in one-to-one correspondence with the outcome classes the screening protocol accepts: core, peripheral, skin, tympanic and axillary temperature; thermogenesis, diet-induced thermogenesis and thermic effect of food; resting and postprandial energy expenditure and metabolic rate; peripheral circulation, blood flow and microcirculation; thermoregulation and cold tolerance; and subjective thermal sensation and cold sensitivity. An earlier version of this analysis used four terms (*thermogenesis, body temperature, peripheral circulation, thermoregulation*), which is narrower than the rule the screening then applies — and a query cannot measure the absence of evidence about an outcome class it never searches for. That narrower set demonstrably missed on-construct studies (see below), so it was replaced and all 175 queryable foods were re-queried on 2026-08-08; both vocabularies and both count snapshots are published. Terms known to add noise were kept deliberately: plant “cold tolerance” pulls in agronomy records and “blood flow” pulls in veterinary ones, but both concepts sit inside the inclusion rule, and screening can remove noise whereas nothing downstream can recover a record the query never returned. Japan-specific foods with weak English coverage were given minimal English synonyms (for example, natto → “fermented soybean”); the inclusion criterion was held constant across foods so that counts are comparable. Queries are built deterministically from each food’s key, and the raw count response for every query is published, so each reported count is pinned to its retrieval snapshot.

Why the raw count is not the measure. L2 counts keyword co-occurrence, not studies of the construct, and it fails in both directions. It over-counts heavily: chicken returned 702 hits dominated by avian thermoregulation and poultry heat-stress research, 566 of them (80.6%) animal studies. An analysis built on L2 would report the research attention paid to a keyword pattern, not to the belief. Screening addresses that failure. The opposite failure cannot be fixed downstream, because screening is a filter and L2’ is a subset of L2: it removes false positives but never recovers a study the query did not retrieve. This is why the effect vocabulary was rebuilt rather than left at four terms, and the case that forced it is concrete — Fagundes et al. (2021), a randomized double-blind crossover trial of ginger extract measuring thermic effect of food by indirect calorimetry alongside axillary temperature, was returned by none of the four terms. Under the 22-term query it is retrieved, and both coders independently labelled it include. Residual under-counting — evidence outside PubMed, or reported in vocabulary no term of ours captures — is carried into the Limitations as a floor on the measure.

Screening. We retrieved the title, abstract, journal, and publication types for every L2 hit and screened all **12,437 records across 138 foods** to a single construct: *a study of the thermal effect, on a human, of ingesting the food or its principal dietary constituent*. The include/exclude definition is documented in `data/screening_protocol.md` and fixed before coding. Include requires all of: human subjects in

vivo; oral ingestion of the food, a food form of it, or its principal dietary constituent; a thermal-related physiological outcome (core, peripheral, skin, tympanic or axillary temperature; thermogenesis or thermic effect of food; energy expenditure reported in a thermogenic context; peripheral circulation; thermoregulation or cold tolerance; subjective thermal sensation); and attribution of the outcome to the food as the exposure of interest. Exclusions are coded by reason (animal, livestock-heat, invitro, agri, name-only, no-thermal, mechanism-only, not-ingestion, species-mismatch, and the two information-shortfall flags *uncertain-species* and *no-abstract*, which route a record to the author regardless of whether the coders agreed). Isolated constituents given at dietary doses (caffeine for coffee, capsaicin for chili pepper) are included and sub-labelled; a different species in the same family is excluded, so that *Aframomum* and *Kaempferia* studies do not inflate “ginger.” No quality grading was performed — this is construct-validity screening, not risk-of-bias assessment. The 37 queried foods with zero L2 hits have $L2' = 0$ by construction and required no screening; foods queried but not screened would be recorded as missing rather than zero, and there were none.

Three boundary questions arose during screening and were settled in the protocol before the affected records were judged, so that the same rule applied across every food. First, the *vascular bed*: “peripheral circulation” reaches cutaneous and limb perfusion — skin blood flow, forearm, calf, hand and foot flow, extremity rewarming — but not organ-specific perfusion measured for a non-thermal purpose (cerebral, coronary, renal, hepatic, retinal, ocular, sublingual, placental, splanchnic, or skeletal-muscle perfusion for nutrient delivery). Without that line, any vascular-function trial of any food becomes an include and $L2'$ fills with endothelial-function literature the folk claim never addressed. Second, *energy expenditure* qualifies where the paper frames it as thermogenesis, diet-induced thermogenesis, thermic effect of food or the food’s thermogenic effect, and not where it appears as a covariate, as the denominator of an exercise energy balance, or as a body-composition or weight-loss endpoint. Third, *reactivity probes* — acetylcholine, sodium nitroprusside, methacholine, insulin-stimulated limb flow, flow-mediated dilatation, post-occlusive reactive hyperaemia — read in a hypertension or endothelial-function framing are endothelial endpoints rather than thermal ones, whereas **local thermal hyperaemia**, the skin’s perfusion response to local heating, is thermal by construction and qualifies.

Screening reliability. Every record was labelled independently by two large language models of **different capability tiers from one vendor’s line**: Claude Sonnet 4.6 with Claude Opus 4.7 for the 623 records of the five golden-set foods, and Claude Sonnet 5 with Claude Opus 5 for the 2,264 records of the remaining foods in the four-term set and for the 9,550 records the 22-term query added. Model versions differ across batches because screening ran over several working sessions; within each batch the two coders are contemporaneous. The recall-rebuild batch deliberately reuses the same two tiers as the batches carrying most of the existing judgments, so the widened record set is judged by the same instrument as the set it extends, and the 2,887 judgments made under the four-term query are reused verbatim rather than re-elicited — re-coding a record already judged would put two labels on one key. We state the pairing precisely because it bounds what the agreement statistic can mean: two models from one vendor and one release generation share pretraining data, tokenizer and alignment methodology, so their errors should be

expected to correlate, and their agreement measures the **consistency** of that lineage rather than the convergence of independent judgments. This is a weaker design than the three-vendor arrangement of Hilkenmeier et al. (2026), and we do not claim its properties. Each agent received the protocol and its assigned records only — the other coder’s labels, the golden set, and the reconciliation code were withheld. Cohen’s κ was **0.806 on all 12,437 co-coded records** (observed agreement 0.986), with 174 divergences; it was 0.811 on the 2,887 records of the four-term set alone, so widening the vocabulary did not destabilise how the protocol was applied. Both coders returned every record, so Axis B has none of the coverage differences that Axis A had to adjudicate, and κ ’s denominator here is the entire record set rather than an agreed subset of it. The author adjudicated **336 records** against title and abstract, in two passes. The first settled 210: every divergence, plus every record either coder flagged as not stating the subject species or as having no abstract, since agreement reached on information a record does not contain is not evidence about that information.

The second pass exists because coder agreement is not self-certifying. Two coders who agree on a wrong include are never routed to adjudication, and the three boundary rules above were written down after the coding pass had run, so agreed includes could not have been judged against them. The author therefore re-read **all 380 agreed includes** against each record’s own title and abstract rather than against the coders’ stated reasons: **115 are excluded** under the vascular-bed, energy-expenditure and reactivity-probe rules, and 11 flagged by a keyword scan are confirmed as includes. A keyword scan alone was not sufficient and is reported as inadequate rather than used: it mis-sorted retinal microvascular density and optic-nerve-head microcirculation into the cutaneous tier on the word “microcirculation,” and it flagged two records whose outcome is local thermal hyperaemia and which are therefore correct includes. Every override is filed in the published ledger with its rationale and appears in `screening.csv` as author-adjudicated, so no label changes silently. Before full screening, the author hand-labelled a 169-record golden set (29 include, 140 exclude) spanning the foods where the include boundary actually lives; against it, coder 1 achieved precision 0.867 / recall 0.897 / F1 0.881 and coder 2 precision 0.813 / recall 0.897 / F1 0.853. Gold labels for records whose abstract does not state the subject species were set from PubMed Humans/Animals Medical Subject Headings (MeSH) or full text, never by inference; this check corrected one gold label, and the correction was surfaced by a coder divergence rather than by the author’s own review.

A third, adversarial pass over the foods that could flip. Two coders who exclude the same record for the same wrong reason leave no trace in κ , and because the primary outcome is binary, one such record would change a food’s result outright. We therefore re-screened the records that stood to flip: all **419 candidates belonging to the 28 core foods (≥ 3 Tier-1 sources) whose L2’ is zero** and whose query returned anything at all. A third agent (Claude Opus 5), blind to the existing labels, to the golden set and to the reconciliation code, was told the opposite of what the first two coders were told — that every record it was about to read had been excluded, and that its task was to find the ones where that was wrong. Its verdicts were `exclude-agreed`, `uncertain`, or `include-candidate`, with standing instructions to return `uncertain` whenever it hesitated and never to infer an unstated subject species. It returned **no include candidates and five uncertain**s; the author read all five against title, abstract and MEDLINE

indexing, and **none changed label**. Every verdict and every ruling is published in `data/screening_thirdpass.csv`. The nine remaining core-zero foods returned no L2 hits at all and so have nothing to re-screen; their zero is a question about the search vocabulary rather than about screening, which is where the Limitations place it.

No publication-year restriction was applied at any layer; the retrieved records span the whole of PubMed's coverage, the oldest included here dating from the 1950s.

Auxiliary databases. OpenAlex (keyword-based) and the CiNii Research API (Japanese-language literature) were queried as auxiliary columns. OpenAlex `search=` is a full-text match and returns counts an order of magnitude larger than PubMed `[tiab]`, so the two are reported separately and never summed. The CiNii query retrieves the Japanese food name alone — an L1-equivalent total, **not** a thermal-context count; it therefore bounds how much Japanese-language literature exists on each food but does not establish that a food's thermal claim is unstudied in Japanese. The primary axis is the screened PubMed count.

Analysis

The analysis plan was fixed before any estimate was computed and is recorded in the project PLAN. The **primary model** is a logistic regression of whether a food has any on-construct study ($L2' > 0$) on lay-source coverage (`n_sources`), adjusted for $\log(L1 + 1)$; the unadjusted fit is reported beside it. A binary outcome is what the data can support — $L2'$ is sparse, with most foods at zero and most non-zero foods at one or two studies — and “which foods have no direct research” is the paper's question. **Reported alongside** are Spearman rank correlations of coverage against $L2'$, against raw $L2$, and against $L1$, so that the effect of screening and the role of literature volume are both legible. The **sensitivity model** is a negative binomial regression of $L2'$ with $\log(L1 + 1)$ as an offset, fitted after a Poisson over-dispersion check (Green 2021; the model class has bibliometric precedent in Millones-Gómez et al. 2023). Branch conditions were fixed in advance: if the negative binomial failed to converge it would be reported as such rather than replaced by a zero-inflated model, which this many zeros over this few foods cannot identify; and a null primary result would be reported as no detected association rather than treated as a failed study.

Five analyses are reported that were **not** in the fixed plan and are labelled as such wherever they appear: the rank correlation of coverage against the $L2'/L1$ ratio; the refit of the primary model under the Tier 1+2 frame; and three refits under narrower readings of what counts as an on-construct study — one dropping studies of a food's principal dietary constituent, one dropping reviews and meta-analyses, and one dropping both. The last three were added in revision. Each of those inclusions is defensible for “has this claim been examined at all” and each is arguable, so asserting either would leave the primary count resting on an untested choice. All five were computed after the primary estimates and none is used to support a conclusion.

The zero-research rate is reported at every level of lay-source coverage rather than at selected thresholds, so that no cut-point is chosen after seeing the data. Warming and cooling foods are compared on the zero-research contrast (Fisher exact) and on $L2'$ (Mann–Whitney U) within the core set. All tests are two-sided

at $\alpha = 0.05$; only the primary logistic model is confirmatory, and the remaining comparisons are descriptive and reported without correction for multiplicity. Ninety-five-percent confidence intervals use the Fisher z-transformation with the Bonett–Wright variance for rank correlations, **Wald bounds** for regression coefficients, Woolf’s logit method for the odds ratio, and Wilson score intervals for the proportions in Figure 1. In the negative binomial the dispersion parameter is estimated by profiling the likelihood over a grid and then held fixed, so its interval is conditional on that value and is correspondingly narrow.

The two predictors are associated (Spearman $\rho = +0.465$), but not to a degree that destabilises the fit: on the model’s own design matrix their Pearson correlation is $+0.396$ and each predictor carries a variance inflation factor of 1.19 . The rank correlation is what Table 3 reports, but a variance inflation factor is defined from the linear fit between the predictors, so it is computed there rather than by squaring ρ . $\log(L1 + 1)$ rather than raw L1 enters the model, which compresses a four-order-of-magnitude range (29 records for hojicha, 171,272 for milk) into a scale where no single food dominates. Beyond this we report no model diagnostics: with 28 events over two predictors the model is used to describe an association, not to predict, and calibration or discrimination statistics would over-interpret it.

Analyses used Python 3.14.3 with pandas 3.0.1, scipy 1.17.1, numpy 2.4.2, statsmodels 0.14.6, and matplotlib 3.10.8; the pipeline is covered by 111 unit tests (all passing). Every statistic in the Results and in Tables 3, 4 and 5, and the composition counts in Table 1, are recomputed from the pipeline output by `src/verify_stats.py`; Table 1’s Axis A reliability figures come from `src/reconcile_coders.py`; the screening quantities in Table 2 — the κ , the agreement and adjudication counts, the golden-set scores and the exclusion-reason breakdown — are recomputed by `src/build_screening.py` from the published ledger, and the third-pass row by `src/build_thirdpass.py`. Figure labels are English-only to avoid CJK-glyph problems in the PDF build (weasyprint 68.1).

Results

Sample and the effect of screening

Axis A coding covered 196 distinct foods across 15 sources. Under the primary Tier-1 frame, 146 queryable foods carry at least one source attribution: 51 reach three or more sources (the *core* set), 28 have two, and 67 have one (Table 1). Among the 51 core foods, 31 are lay-classified warming, 19 cooling, and one contested.

Screening changed the measure substantially. Across the 146 foods in the primary frame, 11,715 raw L2 hits reduced to **283 on-construct studies** — a 97.6% reduction. Over the full 175-food query universe, on which the screening was carried out, 12,437 records reduced to 291 (Table 2), of which 264 are primary reports and 27 are reviews or meta-analyses of human thermal-ingestion evidence. Well over half of the

12,146 exclusions are animal research: 6,252 records excluded as animal and a further 858 as livestock-heat, together 58.5%. The next largest classes are name-only (2,206 records, 18.2%), covering studies where the food term matched but nothing about ingestion and body temperature was at issue, and agri (1,428 records, 11.8%) — crop cold tolerance and post-harvest physiology, which is the noise the wider vocabulary was accepted in order to buy recall.

The screening behaved as intended on the foods that motivated it. **Chicken fell from 702 raw hits to five**, a 99.3% reduction, with 566 of the 702 excluded as animal research. Of the five that survive, three concern chicken-derived preparations rather than the meat — an essence tablet, a hydrolysed sternal-cartilage extract, and a multi-ingredient supplement in which chicken essence is one component — leaving two trials of people eating chicken meat, both of nutritionally enriched meat assessed for microvascular reactivity. Lamb shows the pattern in its pure form (620 → 0). Ginger illustrates the opposite failure: 70 raw hits reduced to 12, and those 12 include the thermic-effect and energy-expenditure trials — Mansour et al. (2012), which the raw count had buried, and Fagundes et al. (2021), which the four-term query had not returned at all. Screening surfaces such studies; it cannot retrieve them, which is why the retrieval step had to be rebuilt rather than compensated for downstream.

The adversarial third pass over the 419 candidate records behind the core-zero foods as they stood under the four-term query — 28 of the 37 zero foods then had any record to re-read — returned no include candidates and five uncertain, none of which survived the author's reading (Methods). No food changed outcome. That pass therefore covers the earlier zero set rather than the 25 of Table 4; what covers the current ledger in the opposite direction is the author's sweep of every agreed include, which did overturn 115 records. Between them the two passes address the failure mode their agreement statistic cannot detect — two coders making the same mistake in the same direction — from both sides, though neither eliminates it, since all three agents come from one vendor's line.

Lay-source coverage does not predict whether a food has been studied

Of the 146 foods, **97 (66.4%) have no on-construct study**, and among the 51 core foods the rate is 25/51 (49.0%). Throughout what follows, “no on-construct study” is shorthand for none that this query retrieved and this screen retained. That is a floor on the truth rather than a census of it, for reasons the Limitations set out.

In the planned primary model, literature volume predicted whether a food had any on-construct study — **odds ratio (OR) 2.106 per unit of $\log(L1 + 1)$, 95% confidence interval (CI) 1.583–2.803, $p < 0.001$** — and lay-source coverage did not: **OR 1.038 per additional source, 95% CI 0.854–1.263, $p = 0.705$** ; model pseudo- $R^2 = 0.285$ (Table 3). Unadjusted, coverage *is* associated with having been studied (OR 1.272, 95% CI 1.075–1.504, $p = 0.005$; pseudo- $R^2 = 0.044$). The association is therefore present in the data and is accounted for once literature volume is in the model, rather than being absent from the data to begin with — a distinction the earlier four-term version of this analysis could not draw, because there the unadjusted estimate did not reach significance either.

The coverage estimate is a non-finding rather than a demonstrated null. Compounded across the observed range of one to nine sources, its interval spans an odds ratio from 0.21 to 6.00 — that is, the data are compatible with widely covered foods being either substantially less or substantially more likely to have been studied. What can be said is that no association was detected, and that the same model with the same 28 events detected a clear one for literature volume.

The rank correlations show why adjustment matters. Coverage correlates positively with the screened count ($\rho = +0.305$, 95% CI +0.146 to +0.448, $p < 0.001$) — but it also correlates, more strongly, with the food's total literature ($\rho = +0.465$, 95% CI +0.320 to +0.589, $p < 0.001$). Widely covered foods are, on the whole, foods with large general literatures; once that is accounted for, the coverage association is not distinguishable from zero. Figure 1 shows this directly: the observed proportion studied rises with coverage, from 0.18 at one source to 0.62 at five or six (panel a), and rises monotonically with literature volume, from 0.08 to 0.69 across L1 tertiles (panel b), and when the primary model holds literature volume fixed the coverage curves are flat at every volume level while remaining widely separated from one another (panel c).

The correlation with raw, unscreened L2 was substantially higher ($\rho = +0.381$, $p < 0.001$) than with the screened count, indicating that roughly half of the apparent coverage–attention association in the unscreened data was carried by records that do not measure the construct.

The zero-research rate falls as coverage widens and then stops falling. It is 66.4% across all 146 foods, 53.2% at two or more sources, 49.0% at three or more, and 47.4% at four or more; from there it flattens and rebounds on shrinking denominators — 50.0% at five or more, 57.9% at six or more, 63.6% at seven or more (Table 3). The confidence intervals in Figure 1a overlap from three sources upward, so the descent is carried by the move off the single-source level rather than by widening coverage within the core set. The per-food picture behind these rates is in Figure 2, where the 97 zero-study foods sit on the baseline across the whole coverage range rather than clustering at its narrow end.

The sensitivity model reached the same verdict of non-significance, and now a similar estimate. Poisson deviance/df was 4.719, confirming over-dispersion; the negative binomial with $\log(L1 + 1)$ as offset converged (dispersion parameter $\alpha = 3.384$) with a coverage incidence rate ratio (IRR) of 1.049 (95% CI 0.880–1.252, $p = 0.591$). In the four-term version of this analysis the offset model's coefficient was an order of magnitude larger than the primary model's and its interval nearly excluded unity; the two specifications now agree in estimate as well as in verdict. The offset still constrains the volume coefficient to exactly 1, a constraint the primary model's own estimate (0.745 on the log-odds scale) does not support — which is part of why the offset specification is reported as a sensitivity check rather than as the primary analysis.

Refitting the primary model under the wider Tier 1+2 frame (175 foods, an unplanned check) gave the same answer: 123/175 (70.3%) with no on-construct study, coverage OR 1.042 (95% CI 0.924–1.176, $p = 0.500$).

Narrowing what counts as an on-construct study did not change it either (both checks unplanned). One hundred and eight of the 283 studies examine a food's principal dietary constituent rather than the food itself; dropping them leaves 180 studies across 43 foods, a zero rate of 70.5%, coverage OR 1.023 (95% CI 0.842–1.242, $p = 0.822$) and volume OR 1.955 (1.491–2.565, $p < 0.001$). Dropping instead the 27 reviews and meta-analyses leaves 256 studies across 48 foods (67.1% zero), coverage OR 1.040 (0.856–1.264, $p = 0.693$). Dropping both — the two classes overlap in nine records — leaves 162 studies across 42 foods (71.2% zero), coverage OR 1.024 (0.843–1.244, $p = 0.810$) and volume OR 1.942 (1.481–2.546, $p < 0.001$). The narrowing is not spread evenly: it bites hardest on the foods carrying the most evidence, with green tea falling from 46 studies to 13 and coffee from 30 to 9, since both are substantially carried by constituent trials (Table 5). The estimates do not move.

Widely covered foods with no retrieved study

Twenty-five of the 51 core foods have no on-construct study (Table 4). They include the foods most consistently presented as warming or cooling: carrot, cucumber, and pumpkin at eight of nine Tier-1 sources; burdock, eggplant, mango, and miso at seven. Their total literatures differ by more than two orders of magnitude — hojicha has 29 PubMed records and cucumber 12,174 — yet at neither extreme did the search retrieve a study of the thermal claim. Widening the effect vocabulary moved twelve foods out of this set, tomato and apple among them.

Chicken is the sharpest case, and its shape changed when the query was widened. Six Tier-1 sources present it as warming; PubMed holds 94,310 records on it and 702 in the thermal keyword context; five of those 702 survive screening, and only two examine people eating chicken meat — both trials of nutritionally enriched meat assessed for microvascular reactivity, neither designed to ask whether chicken warms the body. Under the four-term query this count was zero and would have been reported as an absence. The wider vocabulary replaces the absence with a disproportion: roughly 94,000 records on the food, and two on people eating it in a thermal frame.

The asymmetry is in study design rather than in existence. Four of those 702 records put this very question to animals: the postprandial thermic effect of chicken in rats and its thyroid-hormone mediation (Wakamatsu et al. 2013), diet-induced thermogenesis across meat proteins in rats (Ezoe et al. 2016), diet-induced thermogenesis in broilers (Swennen et al. 2006), and — closest of all — a behavioural assay in which mice select their ambient temperature after eating different meats (Wakamatsu et al. 2025), which is the folk construct itself transposed to a rodent. All four are excluded here as animal research, correctly under the protocol, and each is identifiable by its exclusion reason in the published ledger. So the claim has been put to animals in designs built to ask it. In humans, the one retrieved study that set out to measure a thermogenic response to chicken used an essence tablet rather than the meat, and the two studies of people eating the meat measured microvascular reactivity, with local thermal hyperaemia among several vascular outcomes. That is a weaker statement than “the claim has not been put to humans,” which is what the four-term query had licensed; it is also the statement the data support.

Warming versus cooling foods

Warming and cooling foods did not differ. Among core foods the zero-research rate was 22/31 (71.0%) for warming and 14/19 (73.7%) for cooling (Fisher exact OR 0.873, 95% CI 0.242–3.147, $p = 1.000$), and the screened counts did not differ (Mann–Whitney $U = 290.5$, $p = 0.930$; median zero in both groups). We make no claim of a warm/cool asymmetry.

Flagship foods

A handful of foods illustrate that “attention present” does not resolve the question (Table 5). **Green tea** (cooling, four sources) carries the most on-construct evidence in the whole frame with 46 studies, and **white sugar** (cooling, six sources) the second most with 37 — 0.30% and 0.04% respectively of their literatures — but 28 of green tea’s 46 test a constituent and 17 are multi-ingredient formulations, so only 13 survive the narrowest reading. **Coffee** (cooling, four sources) has 30 on-construct studies from 23,286 records, yet 20 of them test caffeine or chlorogenic acid rather than coffee, and the best-powered relevant synthesis reports that caffeine *raises* core temperature under heat exposure (Peel et al. 2025), opposite to the lay attribution. **Ginger** (warming, eight sources) has 12, nine of them whole-food, and those studies disagree: a thermic-effect difference at $p = 0.049$ in Mansour et al. (2012) alongside null total energy expenditure in the same trial, and null in Fagundes et al. (2021) — a trial that the four-term vocabulary had not retrieved at all. Even where the claimed context has been examined, it has been examined thinly, sometimes inconsistently, and in one case in the direction opposite to the belief.

Discussion

Across 146 foods that Japanese lay-facing sources classify as warming or cooling, how widely a food’s attribution circulates showed no detectable relationship with whether that attribution has been directly examined. What was associated with examination was how much the food is studied for any reason: a unit increase in log total literature corresponded to about three-quarters higher odds of having an on-construct study, while an additional lay source corresponded to an estimated 1.5% change with an interval spanning both directions. The positive association visible in the unadjusted data was substantially attenuated once general literature volume was held fixed, which is what one expects if the same foods tend to be both widely written about and widely researched — though, as the Results note, the adjusted interval is wide enough that a real coverage effect of either sign remains possible.

This is a more specific reading than the one we set out to test, and it came from taking the measurement problem seriously. An earlier version of this analysis used the raw keyword co-occurrence count and reported a near-zero correlation between coverage and attention. That correlation was not interpretable, because well over half of the records screened — 7,110 of 12,437 — are studies of animals rather than of people eating food. After screening, the correlation is positive rather than null, and then is no longer distinguishable from zero once literature volume is held fixed, for a reason the raw analysis could not have identified. Construct validity was not a caveat on the result here; it determined what the result was.

The same applies one level down, to decisions the screening protocol had to make inside the construct. Whether a caffeine trial counts as evidence about coffee, and whether a review counts as an examination, are construct judgments rather than statistical ones. We tested both rather than asserting either, and neither moved the estimate. What the test did expose is that the narrowing falls unevenly: green tea carries the largest screened count in the frame and retains nothing under the strictest reading, because all of its evidence is about catechins or is second-hand. The aggregate result is robust to these choices; an individual food's count is not, and Table 5 should be read with that in view.

The finding refines rather than contradicts the prior reviews. Ormsby (2021), Namiranian et al. (2021), and Zhou & Xu (2021) each establish that the field holds partial, mechanistically suggestive evidence, so “no evidence” is the wrong description; our contribution is orthogonal — we show that whatever evidence exists is associated with general research salience, with no association to which claims circulate detected once salience is held fixed, and that a standardized search and screen retrieved none at all for two-thirds of the foods that circulate. Zhou & Xu's observation that food-level research is scarce relative to Chinese-medicine research is echoed quantitatively here: 12,437 records that mention a food alongside thermal vocabulary yield 291 studies of a human eating that food. García-Hernández et al. (2023) showed that a hot–cold belief system can be bibliometrically mapped and found domain-specific knowledge gaps in Mexico; our two-axis design extends that from documenting that a system exists to asking, food by food, what its claims' examination is associated with. Nagata et al. (2017) found that Japan's four warm/cool classification lists disagree when applied to cohort data; that internal disunity and the pattern reported here are complementary symptoms of a field elaborated culturally faster than it has been consolidated empirically.

One line of work deserves separate notice, because it is the closest thing to a sustained programme on this construct and it sits entirely inside our exclusions. Wakamatsu and colleagues have pursued the postprandial thermal effect of meat in rodents for over a decade: the thermic effect of chicken and its thyroid-hormone mediation (Wakamatsu et al. 2013), diet-induced thermogenesis across meat proteins (Ezoe et al. 2016), and most recently an assay in which mice choose their ambient temperature after eating different meats and their fractions (Wakamatsu et al. 2025) — a behavioural read-out of precisely the felt warmth the lay claim describes. The question this paper puts to the Japanese lay literature has therefore been asked, carefully and repeatedly, of rodents. What has not followed is the human study, and that is the shape of the gap these two axes describe.

The practical reading is diagnostic rather than causal. Research attention co-varies with a food's overall research salience, so the foods whose claims are least examined tend also to be those with the smallest general literatures: burdock, daikon, and hojicha are widely presented as warming or cooling and carry a few hundred PubMed records or fewer (323, 245 and 29 respectively). If that pattern holds, these claims are unlikely to meet evidence as a by-product of research on those foods for other reasons. Table 4 names 25 such foods with logged, reproducible counts, and they are the concrete targets a direct-verification programme would take up.

An important interpretive caution applies to caffeine. The evidence that caffeine raises core temperature comes from heat-exposure performance settings (sport, military, and industrial contexts) rather than from studies of everyday subjective warm/cool sensation, and physiological core-temperature change is a distinct measure from felt thermal comfort. We therefore treat the coffee reversal as a well-powered example of belief-incongruent physiological evidence in a specific context, not as a refutation of the subjective experience the belief describes.

Finally, the screening step is itself a transferable result — but a narrower one than a multi-model design would give. What we can claim is tractability: screening an entire 12,437-record hit set to a construct, twice over, with every judgment published, is now feasible at a scale that would previously have forced the sampling compromise this study was originally designed around. What we cannot claim is decorrelated error. Our two coders are adjacent capability tiers of one vendor’s line, not the three-vendor arrangement of Hilkenmeier et al. (2026), so their 98.6% agreement ($\kappa = 0.806$) bounds the internal consistency of that lineage rather than establishing independent convergence; comparing it to the human inter-rater $\kappa = 0.46$ reported by Guo et al. (2024) would also be misleading, since κ depends on prevalence and the two tasks differ sharply there — Guo et al. themselves report a prevalence- and bias-adjusted κ of 0.96 for their own comparison. The 174 divergences were where the difficulty concentrated, and one of them exposed an error in the author’s own reference labels: disagreement between models was useful as a flag for hard records even though agreement between them cannot be read as independent confirmation.

The sharper methodological lesson is that agreement is not self-certifying in the other direction either. Of the 380 records both coders included, 115 were wrong under boundary rules that were written down after the coding pass had run — chiefly endothelial-function trials whose vascular outcome is not a thermal one. No agreement statistic could have surfaced them, because the two coders agreed; only an author pass over every agreed include did. Any pipeline that routes only divergences to human adjudication inherits this blind spot, and the cost of closing it scales with the include count rather than with the divergence count.

Limitations

First — lay-source coverage is a source count, not a measure of conviction. Axis A counts how many independent sources publish an attribution; it does not weight by readership, and appearing in many sources is not the same as being widely believed. The frame is 15 Japanese lay-facing web sources frozen in advance and dominated by corporate and association wellness media. A different frame — print encyclopedias of *yakuzen*, or reader surveys — could shift individual foods’ values. Widening the frame to include the six individual-expert blogs did not change the conclusion (an unplanned check; OR 1.006, $p = 0.926$).

Second — the coders were language models, and neither reliability statistic supports an independence claim. All coding on both axes was performed by large language models under the author’s direction; there were no human raters. On Axis A, $\kappa = 1.000$ covers one thing only — the direction assigned to items both coders extracted — and round 2 paired two instances of the same model while round 1 paired two capability tiers of one vendor’s line, so it measures the consistency of that lineage on the easier half of the task. Extraction, meaning which foods to record at all, is where the coders actually diverged: their overlap in round 2 was a Jaccard index of 0.54, and every difference was settled by the author against the source text rather than by any agreement statistic. Axis B is the same arrangement — Sonnet with Opus — so its $\kappa = 0.806$ is a within-vendor consistency estimate too. Models sharing pretraining data and alignment methodology should be expected to share failure modes, and perfect agreement across a tier boundary (Axis A round 1) is evidence of that rather than against it. Neither figure is comparable to a κ between trained human screeners, and κ is in any case prevalence-dependent, which our 2.3% include rate makes acute. What does not depend on model agreement is the verbatim grounding check on every Axis A row, the golden-set precision and recall against hand labels, and the publication of every screening judgment with its PMID, all of which can be audited directly. The failure mode that no agreement statistic can detect — both coders reaching the same wrong label — was addressed from both sides. On the exclude side, a third pass instructed to overturn rather than to concur read all 419 candidate records of the core-zero foods as they stood under the four-term query and overturned nothing; widening the query changed which foods are core-zero (37 down to 25), so that pass covers the earlier set rather than the current one. On the include side, the author’s sweep over all 380 agreed includes overturned 115, which is direct evidence that the failure mode is real rather than hypothetical. Both bound the concern without eliminating it, since every agent here is drawn from the same vendor’s line.

Third — the screened count is a floor, not a census. L2’ counts studies that the 22-term PubMed query retrieved *and* that survived screening. Screening is a filter, so it can only remove; a study phrased entirely in other vocabulary never enters the candidate set and no amount of screening recovers it. That this failure is real rather than hypothetical is demonstrable from this study’s own history. The first version of this analysis used four effect terms, which was narrower than the inclusion rule the screening applied, and it missed Fagundes et al. (2021) — a randomized crossover trial of ginger measuring thermic effect of food, indirect calorimetry and axillary temperature — entirely. Widening the vocabulary to match the inclusion rule retrieved it, and brought in enough further evidence to cut the core-zero set from 37 foods to 25 and the overall zero rate from 80.8% to 66.4%. This also puts a number on the earlier query’s recall. Taking the 291 studies the 22-term query retrieved and screening retained as the reference set, the four-term query had returned **104 of them — a recall of 35.7%** — so 187 on-construct studies were outside its reach, and any absence it reported was an absence at that recall. The correction is instructive precisely because it can recur: the 22 terms are matched to our own inclusion rule, not to the full space of ways a thermal effect can be reported, so L2’ still under-counts by an unknown margin. Zero means “no study was retrieved by this query and retained by this screen,” not “no such study exists.”

Fourth — database coverage is incomplete and the Japanese-language check is weaker than it should be. PubMed does not index J-STAGE/CiNii, KMBASE/OASIS, or CNKI, so the counts under-represent Japanese, Korean traditional-medicine, and Chinese literatures. Our CiNii query returns total counts for the Japanese food name, not thermal-context counts, so it bounds the size of the Japanese literature on each food but cannot confirm that a food’s thermal claim is unstudied in Japanese. A thermal-context search of Japanese-language databases is the most direct extension of this work.

Fifth — search terms are not equally apt for every food. Some synonyms conflate distinctions the belief makes (green tea and black tea share *Camellia sinensis*), and the unit of the query does not always match the unit of the claim (lay sources speak of chicken breast; the query returns work on the whole species). Screening removes the resulting false positives but cannot recover a study the query never retrieved, so residual differences in query aptness remain a source of between-food measurement error.

Sixth — cross-sectional snapshot. Both axes were measured once, in August 2026. Lay sources are edited and PubMed counts grow, so specific values will drift, although the structural pattern is unlikely to reverse on the timescale of source updates.

Seventh — descriptive, ecological, unregistered, and imprecise. The design identifies an association pattern across foods, not a mechanism or a cause, and food-level associations should not be read as statements about individuals. The analysis plan was fixed before estimation but was not registered on a public protocol registry, and five reported analyses were computed outside it and are labelled where they appear. Precision is the binding constraint on the central result: 28 of 146 foods have any on-construct study, and the coverage interval compounds across the observed one-to-nine-source range to an odds ratio between 0.21 and 6.00. The absence of a detected coverage effect is therefore a non-finding, not evidence that no effect exists, and a frame with more sources or a database with more on-construct studies could resolve it either way.

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Ethical considerations

This study used exclusively publicly available information: aggregate bibliographic counts and public bibliographic records from public literature databases (PubMed, CiNii, OpenAlex) and publicly published lay-media web pages. No individual-level, health, or personally identifying data were accessed, and no human subjects were recruited or contacted. Because the analysis is restricted to already-public bibliographic data and public web content, it does not meet the standard threshold for human-subjects research under either domestic (Japanese Ethical Guidelines for Medical and Biological Research Involving Human Subjects, 2021 revision, for research using publicly available information) or international (e.g., 45 CFR §46.102 for non-human-subjects data) frameworks; accordingly, no institutional review board approval or waiver was sought or required. The study was conducted in accordance with the principles of the Declaration of Helsinki where applicable to research using aggregate, publicly available data.

Acknowledgments

Large language models were used as instruments in this study, not only as writing aids, and their roles are separated here accordingly.

As coding instruments. Axis A attributions were coded twice independently by Claude Opus 4.8 and Claude Sonnet 4.6 (Anthropic). Axis B abstract screening was performed twice independently by paired models of adjacent capability tiers — Claude Sonnet 4.6 with Claude Opus 4.7, and Claude Sonnet 5 with Claude Opus 5 — under the protocol in `data/screening_protocol.md`. The adversarial third pass over the core-zero foods was performed by Claude Opus 5. All models are from a single vendor; the Methods and Limitations state what that does and does not license. Every divergence, every record

flagged for missing information, and every coverage difference was adjudicated by the author against the primary record. Model versions differ across batches because the work spanned several sessions; the batch-level assignment is tabulated in the protocol files.

As development and writing aids. Claude Opus 4.8 and Claude Opus 5 (Anthropic) assisted with code drafting, debugging, literature-search support, and manuscript editing.

All data extraction, statistical outputs, and numerical results were independently checked by the author by re-executing the analysis code (Python 3.14.3; unit tests $n = 111$, all passing) and by recomputing every reported statistic directly from the pipeline output before writing. The author is solely responsible for the accuracy of the numerical results, the choice of specifications, the interpretation, and the conclusions. All references were verified against PubMed, CrossRef, and publisher records; DOIs and PMIDs were confirmed against primary source pages, and one cited work (Jin et al. 2025) is not indexed in PubMed and is cited by DOI.

Author Contributions (Contributor Roles Taxonomy, CRediT)

Mizuki Shirai: Conceptualization, Methodology, Investigation, Data Curation, Formal Analysis, Software, Writing — Original Draft, Writing — Review & Editing, Visualization, Project Administration.

Conflict of Interest

The author declares no conflicts of interest per International Committee of Medical Journal Editors (ICMJE) guidelines.

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Data Availability

All input data used in this study are publicly available. Axis B counts derive from the NCBI PubMed E-utilities (<https://www.ncbi.nlm.nih.gov/books/NBK25501/>), the CiNii Research API (https://support.nii.ac.jp/en/cinii/api/api_outline), and OpenAlex (<https://openalex.org/>). Query strings are generated deterministically by the published query builder (`src/food_query_terms.py`, `src/evidence_mapping.py`) from each food's key, so every query in the study can be regenerated exactly; the 875 raw count responses that those queries returned are published under `data/query_log/`, pinning each reported count to its retrieval snapshot. The one class of raw response **not**

redistributed is the PubMed efetch dumps of titles and abstracts, which are third-party content; they are regenerable with `src/fetch_l2_records.py`, and every PMID they contain appears in the published screening ledger.

Axis A source pages are the 15 publicly published lay-media URLs listed in `data/sources.csv` with their access dates; the raw HTML is not redistributed, and each coded row in `data/claims.csv` carries the verbatim supporting quotation, so every attribution can be located in the live source. The full screening ledger — one row per (food, PMID) with both coders’ labels, whether the author adjudicated it, the final label, and the reason code — is published as `data/screening.csv`, the hand-labelled reference set as `data/screening_golden.csv`, and the two coding protocols as `data/coding_protocol.md` and `data/screening_protocol.md`. The adversarial third pass is published as `data/screening_thirdpass.csv`, one row per re-screened record carrying its verdict, its reason, and the author’s ruling where one was required.

Analysis code (Python 3.14.3, pandas 3.0.1, scipy 1.17.1, numpy 2.4.2, statsmodels 0.14.6, matplotlib 3.10.8, weasyprint 68.1) — including the source-fetching and text-extraction scripts, the claim-coding and Axis A aggregation modules, the Axis B query builders and PubMed/CiNii/OpenAlex collectors, the record-fetching and screening modules, the query logs, the figure pipeline, and the four verification scripts that between them recompute every number reported here (`src/verify_stats.py` for the analysis statistics, `src/reconcile_coders.py` for the Axis A agreement figures, `src/build_screening.py` for the screening reliability and exclusion counts, and `src/build_thirdpass.py` for the third-pass ledger) — is publicly available at <https://github.com/rehabilitation-collaboration/warming-cooling-foods> (MIT License). All numerical results reported in this manuscript are reproducible from the analysis code and published ledgers at repository commit 93b51c3, recorded as the reproducibility anchor for this revision; the anchor for the first posting was 83394075d4ab93e60393a9aa8595a78dfe0b2a77, which reproduces the earlier four-term measure rather than the one reported here.

Tables

Table 1. Sample composition

Set	Definition	n foods	Warming	Cooling	Contested/neutral
All coded	≥ 1 source, any tier	196	—	—	—
Tier-1 covered	≥ 1 Tier-1 source	157	—	—	—
Analysed (primary frame)	queryable, ≥ 1 Tier-1 source	146	77	61	8
— core	≥ 3 Tier-1 sources	51	31	19	1
— two-source	2 Tier-1 sources	28	—	—	—

Set	Definition	n foods	Warming	Cooling	Contested/neutral
— single-source	1 Tier-1 source	67	—	—	—

Coding covered 196 distinct foods across 15 sources (Tier 1: 9 corporate/association; Tier 2: 6 individual-expert blogs), 649 rows, two-coder Cohen's $\kappa = 1.000$ for direction on the 481 co-coded items — extraction overlap was lower (Jaccard 0.54 in round 2) and every coverage difference was adjudicated by the author against the source text (Methods). Of the 157 foods with Tier-1 coverage, 11 are composite category labels (“nuts,” “spices,” “shellfish,” “leafy greens,” “citrus,” “red meat and fish,” “chinese tea,” “yellow-green vegetables,” “small fish,” “fatty meat,” “animal fat”) that cannot be turned into a single unambiguous query and were excluded from Axis B. The 29 queried foods with no Tier-1 source lie outside the primary frame and enter only the Tier 1+2 sensitivity refit.

Table 2. Effect of construct screening on the research measure

Quantity	Value
L2 records screened (all queried foods)	12,437
Foods with ≥ 1 L2 hit / queried with none	138 / 37
Two-coder Cohen's κ (co-coded n = 12,437)	0.806
Observed agreement	98.6%
Coder divergences	174
Records adjudicated by the author	336
— divergences and information-gap flags / §2.3 sweep of agreed includes	210 / 126
Records surviving as on-construct (L2')	291
— primary reports / reviews	264 / 27
Raw L2 $\rightarrow$ L2' within the primary frame	11,715 $\rightarrow$ 283 (−97.6%)
Author sweep of agreed includes: records / excluded / confirmed	380 / 115 / 11
Third-pass re-screen of core-zero foods (four-term query): records / flagged / overturned	419 / 5 / 0

Exclusion reasons (n = 12,146):

Reason	n	Share
animal (non-human subjects)	6,252	51.5%
name-only (food named, no thermal-ingestion question)	2,206	18.2%
agri (crop cold tolerance, post-harvest physiology)	1,428	11.8%
livestock-heat (farm-animal heat stress)	858	7.1%
no-thermal (human ingestion, no thermal outcome)	724	6.0%

Reason	n	Share
invitro	313	2.6%
not-ingestion (non-oral administration)	180	1.5%
species-mismatch (different species in the same family)	159	1.3%
other (mechanism-only, uncertain-species)	26	0.2%

L2 is the raw count of PubMed records in which a food term co-occurs with one of the 22 thermal-effect terms; L2' is the subset that screening retained as on-construct. Golden-set validation (169 hand-labelled records: 29 include, 140 exclude): coder 1 precision 0.867 / recall 0.897 / F1 0.881; coder 2 precision 0.813 / recall 0.897 / F1 0.853.

Table 3. Primary and supporting statistics (primary frame, n = 146)

Analysis	Estimate	95% CI	p
Logistic, adjusted — lay-source coverage	OR 1.038	0.854 to 1.263	0.705
Logistic, adjusted — log(L1 + 1)	OR 2.106	1.583 to 2.803	< 0.001
Logistic, unadjusted — coverage	OR 1.272	1.075 to 1.504	0.005
Spearman ρ — coverage vs L2'	+0.305	+0.146 to +0.448	< 0.001
Spearman ρ — coverage vs raw L2	+0.448	+0.300 to +0.574	< 0.001
Spearman ρ — coverage vs L1	+0.465	+0.320 to +0.589	< 0.001
Spearman ρ — coverage vs L2'/L1	+0.271	+0.110 to +0.417	0.001
Negative binomial (offset log(L1+1)) — coverage	IRR 1.049	0.880 to 1.252	0.591
Logistic, adjusted — coverage, whole-food studies only	OR 1.023	0.842 to 1.242	0.822
Logistic, adjusted — coverage, primary reports only	OR 1.040	0.856 to 1.264	0.693
Logistic, adjusted — coverage, both narrowings applied	OR 1.024	0.843 to 1.244	0.810
Fisher exact — warming vs cooling zero rate (core)	OR 0.741	0.236 to 2.329	0.772
Mann-Whitney U — warming vs cooling L2' (core)	U = 309.5	—	0.758

Zero-research rate at each level of lay-source coverage:

Coverage $\geq$	1	2	3	4	5	6	7	8	9
Foods	146	79	51	38	24	19	11	6	1
No on-construct study	97	42	25	18	12	11	7	3	0
%	66.4	53.2	49.0	47.4	50.0	57.9	63.6	50.0	0.0

Primary model pseudo- $R^2 = 0.285$; 49 of 146 foods have ≥ 1 on-construct study. The unadjusted model's pseudo- R^2 is 0.044. Regression intervals are Wald. Poisson deviance/df = 4.719 confirmed over-dispersion before the negative binomial was fitted; its dispersion parameter ($\alpha = 3.384$) was estimated by grid-profiling the likelihood and then held fixed, so the IRR interval is conditional on that value. The single food at coverage ≥ 9 (banana, two on-construct studies) makes that column uninformative on its own; it is shown because the analysis plan commits to reporting every level rather than selected thresholds. Five analyses were computed outside the fixed plan and are labelled as such wherever they appear. The Tier 1+2 refit ($n = 175$) gave coverage OR 1.042 (0.924–1.176), $p = 0.500$, with 123/175 (70.3%) having no on-construct study; the coverage-vs-(L2'/L1) rank correlation on that frame was $p = +0.304$ (+0.163 to +0.433), $p < 0.001$; and the three narrowed-definition refits are the rows above, which drop the 108 studies of a principal dietary constituent, the 27 reviews and meta-analyses, or both (the classes overlap in nine records), leaving 43, 48 and 42 foods respectively with any study (zero rates 70.5%, 67.1% and 71.2%). Volume stayed significant under every narrowing (OR 1.955, 2.085 and 1.942; all $p < 0.001$). Only the adjusted logistic model is confirmatory; the rest are descriptive and uncorrected for multiplicity.

Table 4. The 25 core foods for which the search retrieved no on-construct study, by lay-source coverage

Food	Lay direction	Sources	Total literature (L1)	Raw L2	On-construct (L2')
cucumber	cooling	8	12,174	67	0
carrot	warming	8	5,702	12	0
pumpkin	warming	8	2,631	15	0
mango	cooling	7	3,627	14	0
eggplant	cooling	7	2,072	13	0
miso	warming	7	1,158	15	0
burdock	warming	7	323	0	0
melon	cooling	6	3,898	35	0
watermelon	cooling	6	3,146	31	0
pineapple	cooling	6	2,494	10	0
green onion	warming	6	401	1	0
tofu	cooling	5	1,188	3	0
onion	warming	4	7,727	30	0
tuna	warming	4	3,715	76	0
brown rice	warming	4	2,040	5	0
white rice	cooling	4	1,514	3	0

Food	Lay direction	Sources	Total literature (L1)	Raw L2	On-construct (L2')
kimchi	warming	4	1,180	6	0
daikon	contested	4	245	0	0
sweet potato	warming	3	3,941	15	0
buckwheat	warming	3	2,799	7	0
chinese cabbage	cooling	3	2,115	11	0
prune	warming	3	1,957	10	0
soy sauce	warming	3	1,264	2	0
kiwi	cooling	3	1,195	6	0
hojicha	warming	3	29	1	0

“Raw L2” is the unscreened keyword co-occurrence count. Every one of these 25 foods has at least three independent Tier-1 sources assigning it a thermal direction, and none of the records this query retrieved and this screen retained studies a human ingesting it with a thermal outcome measured. Widening the effect vocabulary from four terms to 22 cut this set from 37 foods to 25, so a zero here is a floor on the evidence rather than a census of it: a study phrased entirely outside even the 22-term vocabulary never entered the candidate set (Limitations, third point).

Table 5. Flagship foods: total volume, keyword hits, and on-construct studies

Food	Lay direction	Sources	L1 total	Raw L2	L2'	Whole-food primary	L2'/L1	What the direct evidence shows
chicken	warming	6	94,310	702	5	3	0.005%	566 of the 702 hits are avian or poultry-production research; of the five that survive, three test chicken-derived preparations and two test people eating enriched chicken meat, both measuring microvascular reactivity
green tea	cooling	4	15,476	256	46	13	0.297%	The largest screened count in the table, but 28 of the 46 test a constituent and 17 are multi-ingredient formulations flagged confounded; 13 survive the narrowest reading
white sugar	cooling	6	85,454	812	37	15	0.043%	Second-largest count and the more widely covered food; a mixture of thermic-effect-of-food trials and taste-stimulus studies recording skin blood flow and temperature

Food	Lay direction	Sources	L1 total	Raw L2	L2'	Whole-food primary	L2'/L1	What the direct evidence shows
coffee	cooling	4	23,286	165	30	9	0.129%	Caffeine <i>raises</i> core temperature under heat exposure (Peel et al. 2025, n = 30, g = 0.44) — opposite to the lay attribution — and 20 of the 30 test caffeine or chlorogenic acid rather than coffee
ginger	warming	8	5,730	70	12	9	0.209%	The most-covered food in the frame and the one with the most whole-food evidence: a thermic-effect difference at p = 0.049 in Mansour et al. (2012, n = 10) — with total resting energy expenditure and the thermic-effect AUC both null at p = 0.43 in the same trial — null in Fagundes et al. (2021, n = 20), and null again for thermoregulatory function on repeated dosing
chili pepper	warming	6	1,943	50	17	9	0.875%	Seven of the 17 are capsaicin or capsinoids at dietary doses; the highest on-construct share among the foods in this table, though not in the frame (mugicha reaches 8.11% on three studies out of 37 records)
milk	cooling	3	171,272	1,932	17	11	0.010%	1,482 of the 1,932 keyword hits are animal or livestock-heat research; the largest total literature in the study and one of its smallest on-construct shares

“Whole-food primary” is an unplanned quantity, added in revision: the count remaining when records sub-labelled as testing a principal dietary constituent and records sub-labelled as reviews or meta-analyses are both removed — the narrowest reading reported in the Results. The two classes overlap, so the column is not L2' minus their sum. L2'/L1 is computed on L2' as reported. Raw L2 and L2' are both on the 22-term query; under the earlier four-term query chicken returned 278 hits and no surviving study.

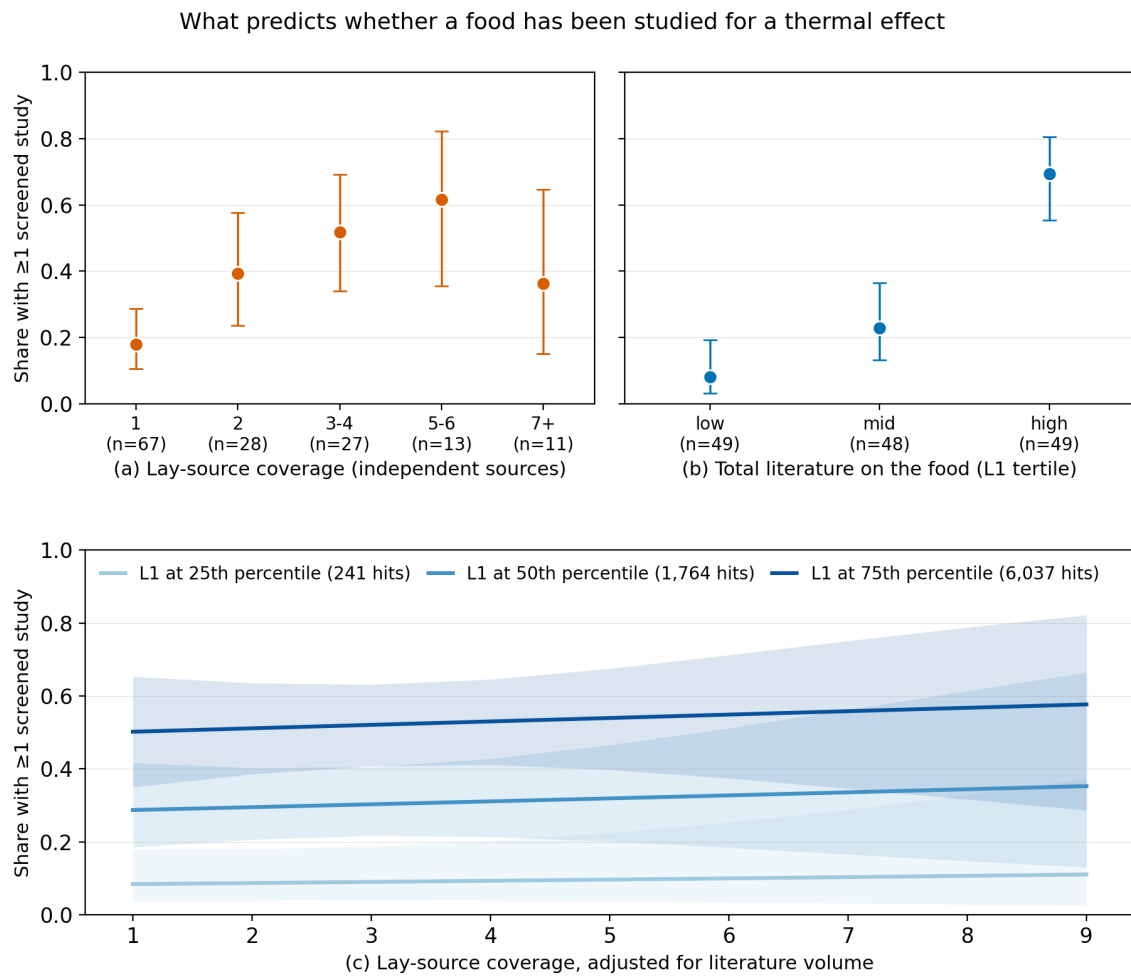


Figure 1. What predicts whether a food has been studied for a thermal effect ($n = 146$). The outcome throughout is the share of foods with at least one on-construct study ($L2' \geq 1$), on a common y axis. **(a)** Observed share by lay-source coverage, binned; group sizes are printed on the axis. **(b)** Observed share by tertile of the food's total PubMed literature (L1). **(c)** The planned primary model: fitted probability across lay-source coverage with literature volume held at its 25th, 50th, and 75th percentiles, with 95% confidence bands. Panels (a) and (b) are marginal and cannot separate the two predictors, because widely covered foods also tend to carry large literatures; panel (c) does separate them — the curves are flat across coverage and widely spaced by volume. Error bars in (a) and (b) are Wilson score intervals.

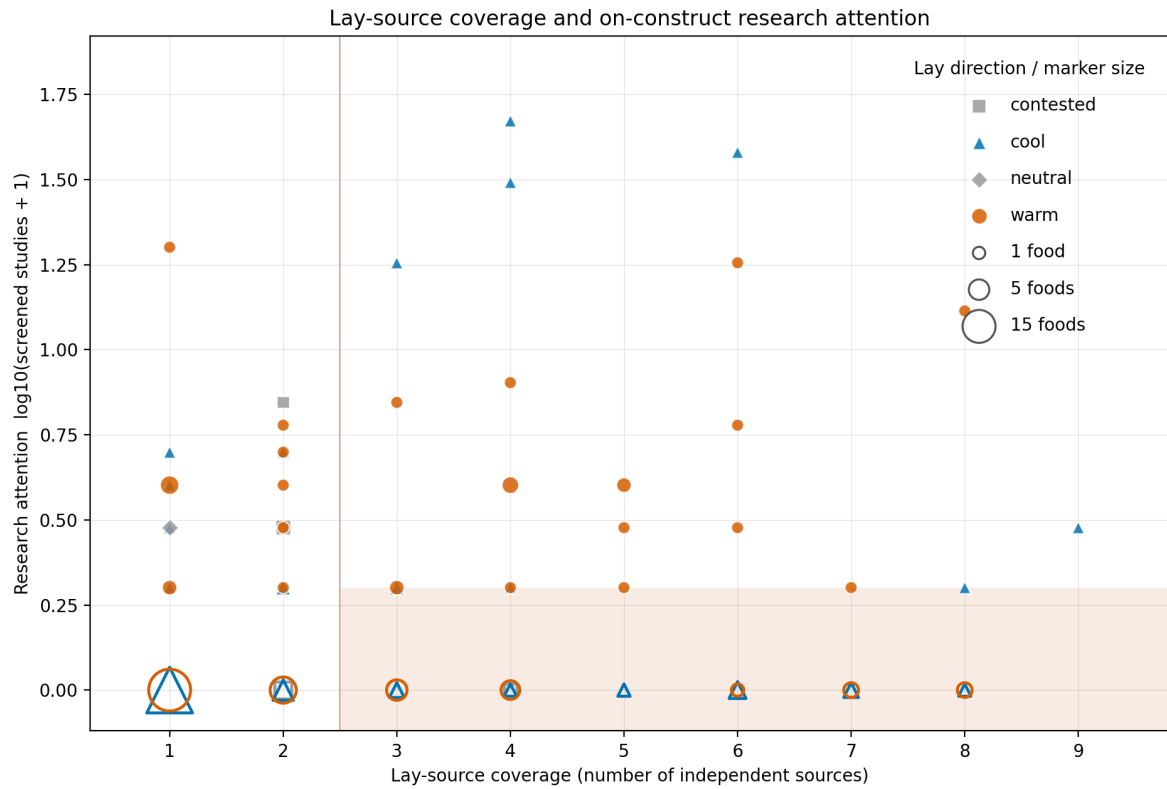


Figure 2. Lay-source coverage against on-construct research attention, per food ($n = 146$). x = number of independent Tier-1 sources assigning a warming or cooling direction; $y = \log_{10}(\text{on-construct studies} + 1)$, so foods with no study sit on the baseline rather than being dropped by the log scale. Lay direction is encoded by both colour and marker shape (warming = vermillion circle, cooling = blue triangle, contested/neutral = grey square/diamond), so identity never rests on colour alone; the Okabe–Ito palette is colour-blind safe. Foods with no on-construct study are drawn hollow, and **marker area encodes how many foods share a position** — 97 of 146 foods sit on the baseline, which a plain scatter would render as a handful of points. The shaded region marks the core set (≥ 3 sources) with at most one study. Individual foods are named in Table 4 and in `plots/attention_gap_data.csv` rather than labelled in place, because leader lines from the baseline would cross studied foods higher up.